



mongo DB

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what is **Mongo Db**?

- **MongoDB** is a cross-platform, document-oriented database providing **high performance, availability, and scalability**
- Works on the concept of **collections and documents**
Designed for **scalability and developer agility**
Built for speed
Supports **rich document-based queries** for easy readability
- **Full index support** for high performance
- **Replication and failover** ensure high availability
Auto-sharding enables easy scalability
- A **shard** is a horizontal partition of data; each shard is stored on a separate server to spread load
MapReduce is supported for powerful data aggregation



MongoDB

Advantages



Performance



Document Model



Flexible Schema

Disadvantages



No Transaction

- No Join



Memory Limitation

MongoDB is NOT RDBMS

- Mongo is not a relational database like SQL Server
- No transactions
- No referential integrity • No joins
- No schema, so no columns or rows (no more ALTER TABLE)
- NoSQL

MongoDB is NOT RDBMS

Great for:

- Large, unstructured or semi-structured data (JSON-like)
- Applications needing fast development & agility
- High scalability (big data, distributed systems)
- Real-time analytics & event logging
- Content management, catalogs, IoT, mobile apps

Not great for:

- Complex multi-document transactions (like banking)
- Strict schema / strong consistency requirements
- Heavy relational data with many joins
- Applications needing strong ACID guarantees at all times
-

RDBMS

Database

Tables

Rows

Columns

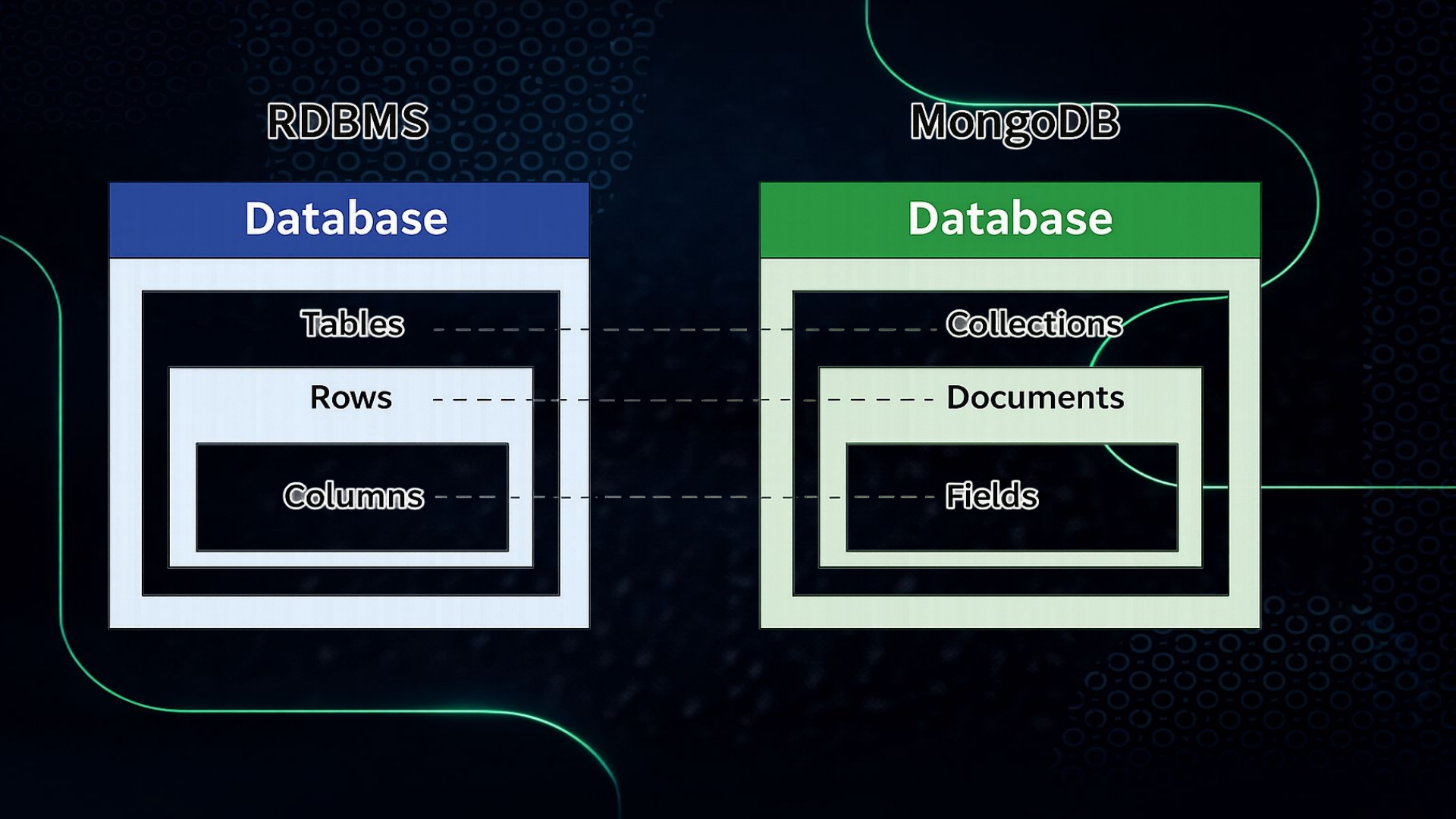
MongoDB

Database

Collections

Documents

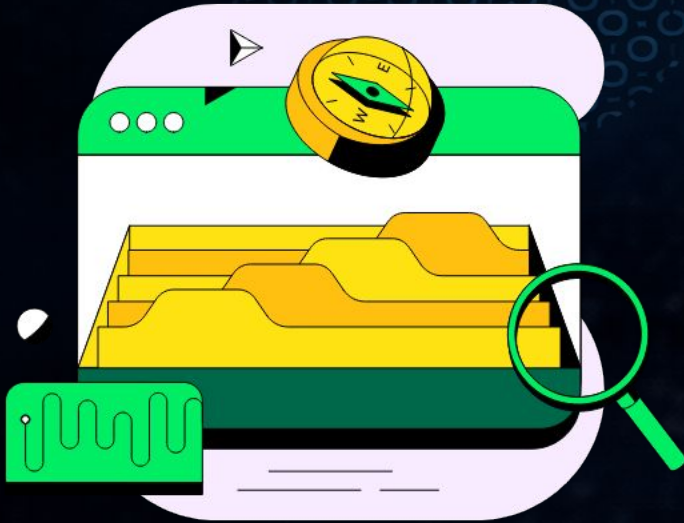
Fields



Agenda

- **CRUD Operations (Create, Read, Update, Delete)**
- **Operators & Query**
- **Projection**
- **Data Modeling (Embedding vs. Referencing)**
- **Aggregation Framework**
- **Backup, Import & Export**
- **MongoDB Atlas (Cloud)**

DOWNLOADS



Mongo Server

<https://www.mongodb.com/try/download/community>

Mongo shell

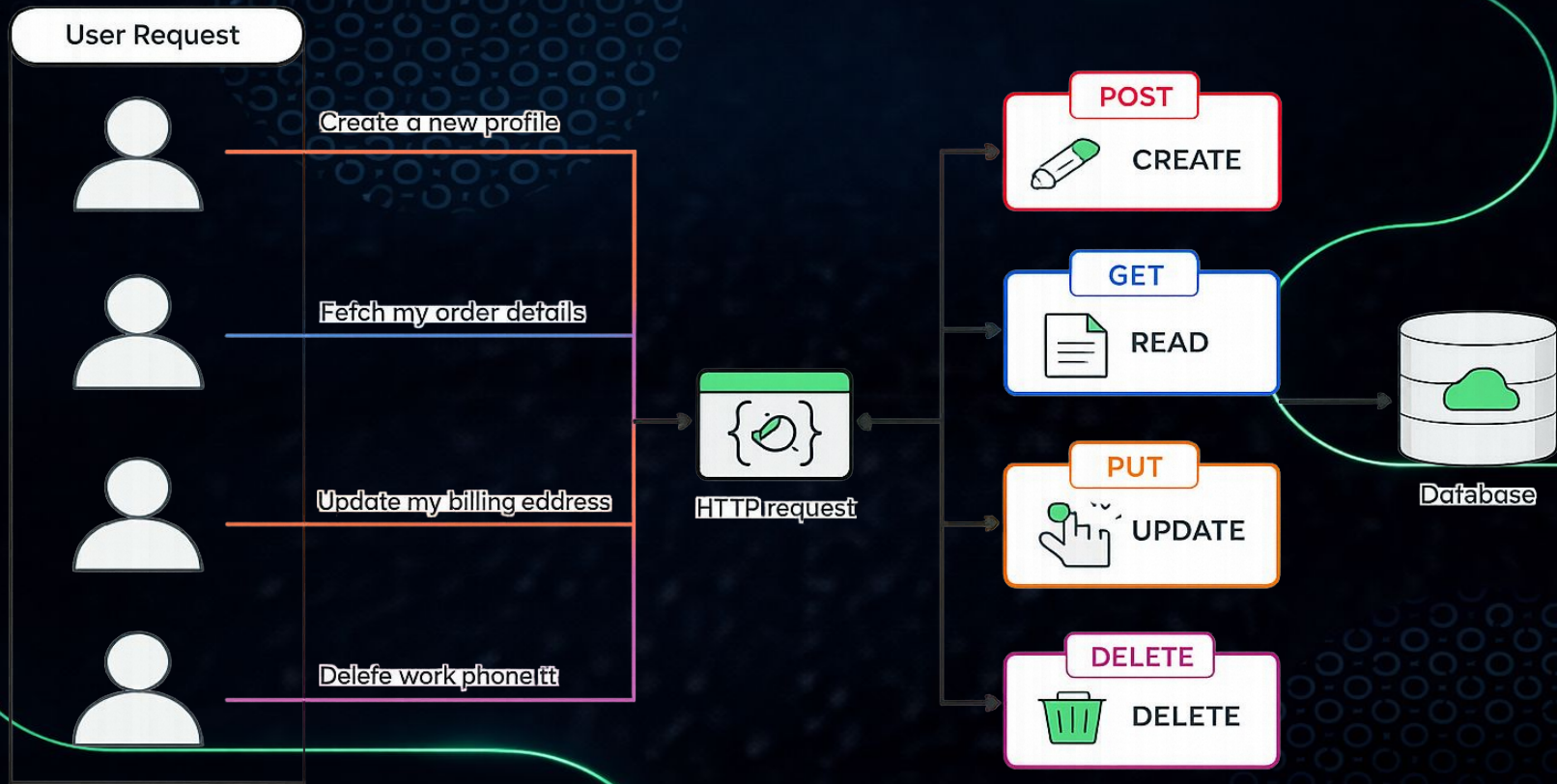
<https://www.mongodb.com/try/download/shell>

Mongo Compass

<https://www.mongodb.com/try/download/compass>

Studio 3T

<https://studio3t.com/download/>





CREATE

- insertOne()
- insertMany()

READ

- find()
- findOne()

UPDATE

- updateOne()
- updateMany()
- replaceOne()

DELETE

- deleteOne()
- deleteMany()

Mongo DB CRUD operation

Comparison

Logical



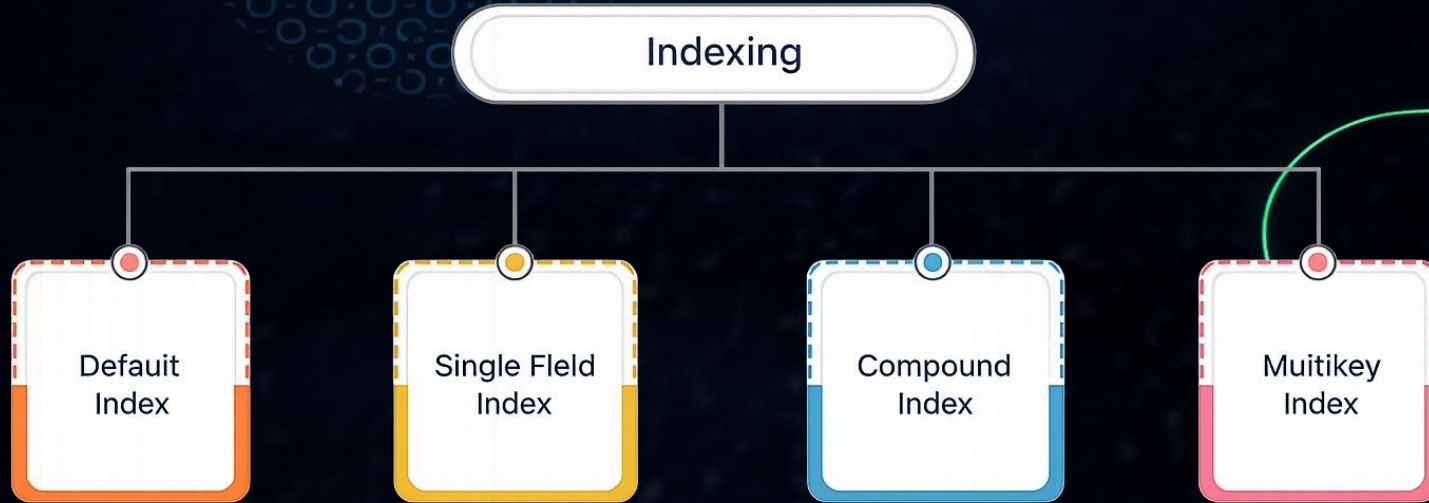
Query Operators

Evaluation

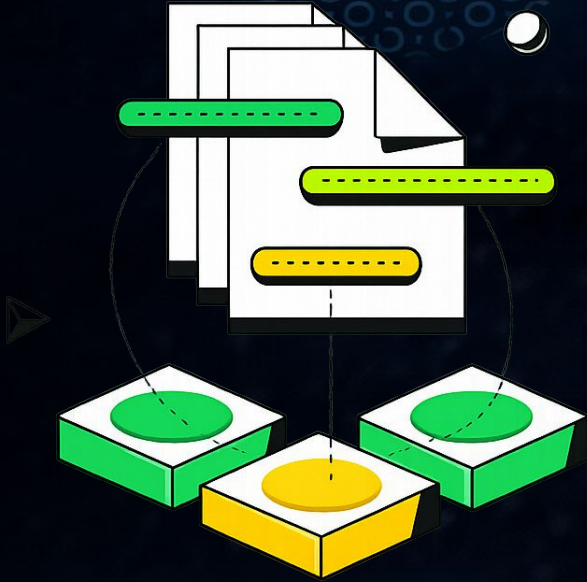
Elements



What is Indexing? Types of Mongo DB Indexings



Data Modeling



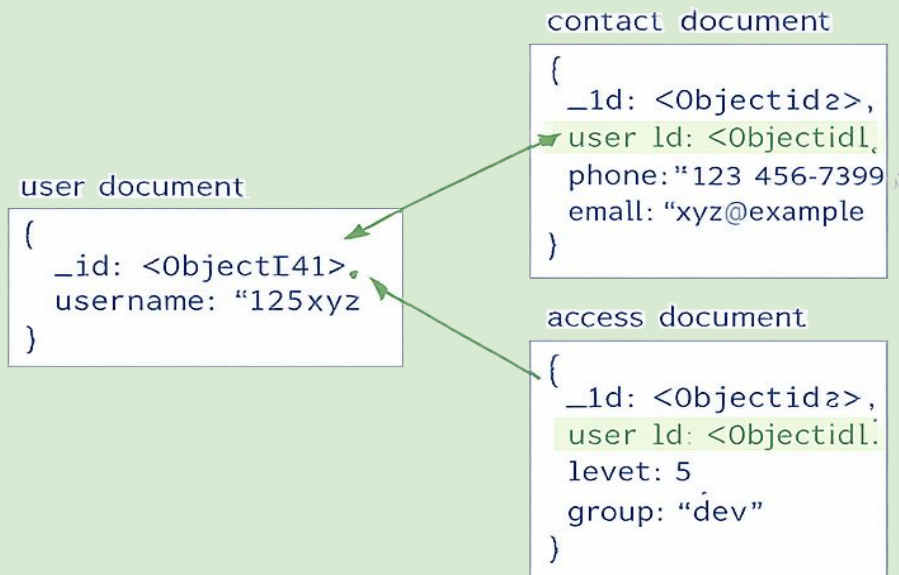
- ❖ The primary function of a data model is to facilitate visual information of how the relationship between two or a group of data points would look.
- ❖ In MongoDB, you can model relationships by either embedding or referencing your data. .

the difference between embedded data models and normalized data models

(a) Embedded Data Model

```
{
  _id: <Objectid1>,
  username: "133yy2",
  contact: {
    phone: "123-456-7590",
    email: "xyz@example.com"
  },
  access: {
    level: 5,
    group: "dev"
  }
}
```

(b) Normalized Data Model



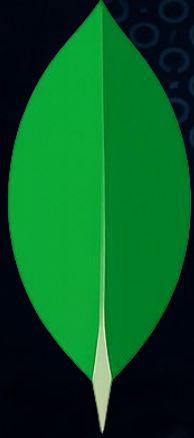
Embedded **vs** normalized

❖ Embedded Data Model

a denormalized data model — is applied when two data sets contain a relationship. Hence an embedded data model sets relationships between data elements, keeping documents in a single document structure.

❖ Normalized data model

uses object references to model relationships between documents. Such a model avoids data duplication: a document can be referenced by many other documents without duplicating its content.



mongoDB

Aggregation Framework



MongoDB Aggregation Pipelines

The MongoDB Aggregation Framework is a powerful tool for performing complex data analysis directly within MongoDB. It allows for **multi-stage data transformation** through an aggregation pipeline, making it ideal for scenarios where you need to process and analyse large datasets.

Aggregation Pipelines in MongoDB consist of **multiple stages** that process data step-by-step, making operations structured, efficient, and scalable. Think of it as a data processing conveyor belt that modifies information at each step!

Collection

db.orders.aggregate(

 Snatch phase → { Snatch: { status: "A" }.

 Sgroup phase → { Sgroup: { _id: "\$cust_id", total: { \$sum: "\$amount" } } })

{ cust_id: "A123", amount: 500, status: "A" }
{ cust_id: "A123", amount: 200, status: "A" }
{ cust_id: "8212", amount: 200, status: "A" }
{ cust_id: "A123", amount: 300, status: "D" }

orders

Snatch →

{ cust_id: "A123", amount: 500, status: "A" }
{ cust_id: "A123", amount: 250, status: "A" }
{ cust_id: "8212", amount: 200, status: "D" }

Sgroup →

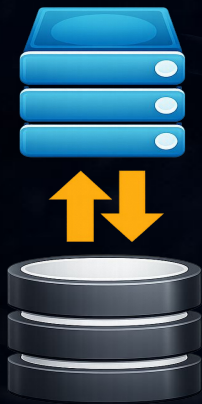
Results

{ id: "A123", total: 750 }
{ id: "8212", total: 200 }

Aggregation Pipelines stages

- ◆ `$match` - Filters documents based on conditions
- ◆ `$project` - Selects specific fields and reshapes the document
- ◆ `$group` - Groups data and performs calculations
- ◆ `$sort` - Sorts results in ascending/descending order
- ◆ `$limit` - Restricts the number of output documents
- ◆ `$lookup` - Performs joins between collections
- ◆ `$unwind`: Deconstructs an array field from the input documents, creating one document for each element in the array

BackUp & Restore, and export & import



Download tools

<https://www.mongodb.com/try/download/data-base-tools>

Mongo tools

mongodump.exe
mongorestore.exe
mongoimport.exe
mongoexport.exe

Command Prompt

>Run as administrator
>cd C:\Program Files\MongoDB\Server\8.0\bin

Documentation

<https://www.mongodb.com/docs/manual/tutorial/backup-and-restore-tools/>

MongoDB Atlas (Cloud)



*Thank
You*

Make free to contact me:

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